Frequency-Bottlenecked CycleGAN: Exposing and Patching Steganographic Vulnerabilities in Unpaired Medical Image Translation

Project Proposal

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1. Introduction

Unpaired image-to-image translation via CycleGAN is a popular approach for medical data augmentation. However, when the two domains are structurally asymmetric (e.g., brain MRI with tumors vs. healthy brain MRI), the standard CycleGAN objective creates a perverse incentive. To minimize cycle-consistency loss without faithfully preserving macroscopic pathology, the forward generator $G : P \to H$ learns to *hide* the spatial coordinates of tumors as imperceptible high-frequency noise in the output. The reverse generator F reads this noise to reconstruct the original, satisfying $F(G(p)) \approx p$ without ever producing a structurally honest translation. This is steganography: the generator is encoding, not translating.

If these synthetic images train downstream diagnostic models, those models learn to detect invisible noise rather than real pathology. This failure mode is documented [1, 2], but existing mitigations rely on heavyweight auxiliary networks (VGG perceptual losses, attention modules) that add millions of parameters and significant training overhead.

2. Proposed Approach: FB-CycleGAN

We propose a lightweight, zero-parameter architectural patch called **Frequency-Bottlenecked CycleGAN (FB-CycleGAN)**. The core idea is simple: inject a fixed, differentiable Gaussian low-pass filter $B(\cdot)$ directly into the cycle-consistency path. By blurring the intermediate generated image before it enters the reverse generator, we destroy the high-frequency bandwidth required for steganographic encoding. The generator is then forced to either preserve pathology in the translated output or accept an irrecoverable cycle-consistency penalty.

2.1 Mathematical Formulation

The standard CycleGAN cycle-consistency loss is:

$$\mathcal{L}_{cyc}(G, F) = \mathbb{E}_p[\|F(G(p)) - p\|_1] + \mathbb{E}_h[\|G(F(h)) - h\|_1] \tag{1}$$

We replace this with the frequency-bottlenecked variant:

$$\mathcal{L}_{\text{FB-cyc}}(G, F) = \mathbb{E}_p[\|F(\,B(G(p))\,) - p\|_1] + \mathbb{E}_h[\|G(\,B(F(h))\,) - h\|_1] \tag{2}$$

where $B(\cdot)$ is a Gaussian blur with fixed kernel size k and standard deviation σ. Both k and σ are hyperparameters, not learned. The full training objective is:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{GAN}}(G, D_H) + \mathcal{L}_{\text{GAN}}(F, D_P) + \lambda\,\mathcal{L}_{\text{FB-cyc}}(G, F) + \gamma\,\mathcal{L}_{\text{id}}(G, F) \tag{3}$$

The blur adds zero learnable parameters. It is differentiable, so gradients still flow through the cycle path, but the information bottleneck prevents high-frequency side-channel encoding.

3. Novelty and Contribution

The steganographic failure mode is known. A minimal, targeted fix that addresses the root cause (high-frequency encoding capacity) without auxiliary networks is not well-studied. We contribute: (1) a controlled empirical demonstration of steganographic hiding under domain asymmetry, verified through spectral forensics (FFT residual maps); (2) FB-CycleGAN, a zero-parameter frequency bottleneck that reduces the information-theoretic capacity of the cycle path; (3) an honesty-vs-quality Pareto analysis across blur strengths σ; and (4) downstream validation showing FB-CycleGAN synthetic data produces better segmentation models.

4. Dataset

We use the **BraTS 2020** training set[1], which contains 369 multi-parametric MRI volumes (293 HGG + 76 LGG) stored as NIfTI files (`.nii.gz`). Each volume comprises four sequences (T1, T1CE, T2, FLAIR) at $240 \times 240 \times 155$ voxels, plus a voxel-level segmentation mask labeling enhancing tumor, edema, and necrotic core. We use the FLAIR sequence (highest tumor-to-background contrast) and extract 2D axial slices via `nibabel`. Slices are assigned to **Domain** P if the segmentation mask shows tumor area $> 5\%$ of the brain region, and to **Domain** H if the mask is entirely zero and brain area exceeds a minimum threshold. Preprocessing per slice: center-crop to brain bounding box, resize to 256×256, single-channel grayscale, Z-score normalization per volume. The segmentation masks are reserved for the downstream U-Net evaluation and are never seen by the CycleGAN.

[1] `https://www.kaggle.com/datasets/awsaf49/brats20-dataset-training-validation`

5. Evaluation Protocol

We train two models on the same data split: a baseline CycleGAN and FB-CycleGAN, identical except for the blur bottleneck.

Steganographic Forensic Audit. For each generated image $\hat{h} = G(p)$, we compute the FFT power spectrum of the residual $|\hat{h} - p|$. Steganographic encoding will appear as structured high-frequency energy in the baseline but not in FB-CycleGAN. We also inject additive Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ before reverse-cycle reconstruction; a steganographic generator will fail catastrophically while a structurally honest one degrades gracefully.

Quality and Fidelity. We report FID for generation quality and SSIM for cycle-reconstruction fidelity. Sweeping $\sigma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ produces a Pareto frontier of the honesty-quality tradeoff.

Downstream Segmentation. A lightweight U-Net trained on synthetic pathological images from each model is evaluated on held-out real BraTS data via Dice score. Higher Dice from FB-CycleGAN data confirms downstream clinical value.

6. Technical Stack

Component	Tool
Framework	PyTorch 2.x + torchvision
Data loading	`nibabel` (NIfTI → NumPy), custom `Dataset` class
Generator arch.	ResNet-9 blocks (modified for 1-ch input)
Discriminator arch.	PatchGAN (70×70 receptive field)
Blur bottleneck	`torchvision.transforms.GaussianBlur`
Spectral analysis	NumPy FFT / `torch.fft`
Downstream model	U-Net (4-level encoder-decoder)
Metrics	`torchmetrics` (FID, SSIM), custom Dice
Training infra	Kaggle / Google Colab (T4/P100 GPUs)
Experiment tracking	Weights & Biases

7. Project Components and Timeline

Week	Component	Deliverable
1	Data pipeline	NIfTI→PNG extraction via `nibabel`, domain filtering, train/val split
1	Baseline CycleGAN	Working 1-channel CycleGAN, initial training run
2	FB-CycleGAN	Blur bottleneck integrated, verified gradient flow
2	Hyperparameter sweep	σ sweep, convergence curves logged to W&B
3	Forensic evaluation	FFT residual maps, perturbation audit, Pareto curve
3	Downstream U-Net	Segmentation trained on synthetic data from both models
4	Ablations + writeup	Final metrics, figures, report, presentation slides

8. Hypotheses

H1: Baseline CycleGAN exhibits structured high-frequency energy in FFT residuals, confirming steganographic encoding. Reverse-cycle reconstruction degrades catastrophically ($> 50\%$ L1 increase) under mild perturbation. **H2:** FB-CycleGAN shows diffuse, unstructured residual spectra and degrades gracefully under perturbation. **H3:** There exists a σ^* achieving both acceptable FID and high SSIM, representing a practical Pareto-optimal operating point. **H4:** A U-Net trained on FB-CycleGAN synthetic data achieves higher Dice on real BraTS test data than one trained on baseline synthetic data.

References

[1] Cohen, J.P., Luck, M., Honari, S. (2018). Distribution matching losses can hallucinate features in medical image translation. *MICCAI*.

[2] Chu, C., Zhmoginov, A., Sandler, M. (2017). CycleGAN, a master of steganography. *NIPS Workshop on Machine Deception*.

[3] Zhu, J.Y., Park, T., Isola, P., Efros, A.A. (2017). Unpaired image-to-image translation using cycle-consistent adversarial networks. *ICCV*.

[4] Menze, B.H. et al. (2015). The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE TMI*. Dataset: `https://www.kaggle.com/datasets/awsaf49/brats20-dataset-training-validation`